Normalising Wavefunction

- Our wavepacket was defined with a Gaussian:

$$\psi(x) = Ne^{-A(x-x_0)^2} e^{ik_0(x-x_0)}$$

- We can normalise it using:

$$\int_{-\infty}^{\infty} \psi^* \psi \, dx = \int_{-\infty}^{\infty} Ne^{-A(x-x_0)^2} e^{-ik_0(x-x_0)} Ne^{-A(x-x_0)^2} e^{ik_0(x-x_0)} dx = 1$$

$$\Rightarrow N^2 \int_{-\infty}^{\infty} e^{-2A(x-x_0)^2} dx = 1$$

- Let $u = x - x_0$, so $du = dx$:

$$\Rightarrow N^2 \int_{-\infty}^{\infty} e^{-2Au^2} du = 1 \Rightarrow N^2 \sqrt{\frac{\pi}{2A}} = 1$$

$$\Rightarrow N^2 = \left(\frac{2A}{\pi}\right)^{1/2} \Rightarrow \boxed{N = \left(\frac{2A}{\pi}\right)^{1/4}}$$